

Spreading 49 Points in the Plane

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Abstract

We place 49 points in the plane so that the squared ratio of the largest to the smallest distance between them is at most 46.97025+. Erich Friedman’s table of best known configurations lists n up to 31. Deleting a point from our 50-point configuration gives 47.93330; this one is smaller. The ratio is computed exactly from the published coordinates. We do not claim it is optimal.

1 Result

Theorem. *There are 49 points in the plane with*

$$\left(\frac{\max d}{\min d}\right)^2 = 46.97025\dots$$

The exact value, a rational number, is given in Appendix A.

2 Context

Erich Friedman’s page *Minimizing the Ratio of Maximum to Minimum Distance* lists the best known configurations for $n \leq 31$. The value for $n = 49$ is not on the page. Deleting a point from our 50-point configuration gives 47.93330; this one is smaller.

3 Verification

The coordinates are decimals with 12 digits after the point, so they are exact rationals; all squared distances, and their largest-to-smallest ratio, are computed with integers. Run

```
python friedman-maxmin2d/verify.py friedman-maxmin2d/coords/n49.txt
```

in <https://github.com/Ge-limin/math719>.

4 How it was found

Random starts of a constrained local optimizer (SLSQP), then basin hopping seeded from the neighbouring sizes (`friedman-maxmin2d/`).

A Exact value

```
numerator:
46970259225479189366740465
denominator:
999999999998128784125557
```

*Found and checked by Claude, a model released by Anthropic, running on the author’s laptop. The author sent the prompts.

B Coordinates

49 points, x and y per line. File: friedman-maxmin2d/coords/n49.txt.

```
1.712001881866 -5.888965153575
-1.196525988086 -8.681025509436
1.934588060080 -3.010449719823
3.341265231754 -4.802912778089
1.877677501730 -4.867800405863
2.914316531015 -3.210779755612
0.765863893613 -8.667059955141
-0.361233904670 -9.230831962093
1.673562720073 -6.888226095892
3.381047243893 -9.114351733151
2.575675888563 -4.151695541046
2.501455862169 -7.449111953505
0.893026902563 -4.693263310781
0.695526112134 -6.679792908681
2.481359271529 -9.550885297019
0.623731281865 -7.677212330161
-0.221486211754 -7.142789848898
-0.211548111912 -8.238136784239
3.674775125329 -3.860166178579
1.527723123083 -9.851847248728
3.471752891327 -7.691028623468
-0.871331927580 -3.757228263742
4.324462683049 -4.620367518584
3.786092130889 -6.741717903802
1.601767889804 -7.885645517373
1.743900501553 -8.875493142352
-1.058049759630 -7.690659751671
4.769289582183 -6.559172644297
0.934680802571 -3.024068691134
-1.981006735919 -7.305756621105
-0.215290913294 -5.147278923981
2.643588473917 -8.438959578485
4.126037824241 -8.447276748466
-1.047655690660 -5.701507105690
3.664927398039 -5.749085490714
-1.156730701374 -6.695540627276
-2.079687677662 -6.310637496710
2.639622700921 -5.515441906565
0.035285401703 -4.179182074300
-0.151497617124 -6.145242057250
4.516721596122 -7.526751794763
0.549686515144 -9.643414061517
1.536505760390 -3.927799425592
4.648124849333 -5.566540231209
-1.970612666949 -5.316603975124
2.815795101730 -6.499801233839
-0.052600543771 -3.183051530351
0.733965273927 -5.680531966364
-1.175798289738 -4.709751352341
```